A LoS-Suppressing Beam Sweeping and Tracking Method Based on Coordinate Descent Algorithm for mm-Wave ISAC System

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***Abstract*—This paper proposes a low-complexity line-of-sight (LoS) path interference suppressing method for real-time beamforming and tracking in unmanned aerial vehicle (UAV) monitoring. The method utilizes a coordinate descent algorithm to alternately estimate and suppress the LoS component during target tracking. Compared with conventional approaches such as extended Kalman filter (EKF) and DNN-based methods which suffer from high computational complexity and reliance on training data, our approach achieves superior real-time tracking probability and accuracy in near-LoS scenarios. Simulation results demonstrate a reduction in mean square error (MSE) and enhanced robustness under low signal-to-noise ratio (SNR) conditions.**

***Index Terms*—LoS path interference, 3-D beamforming, real-time UAV tracking, low complexity, coordinate descent method.**

I. Introduction

Three-dimensional (3-D) beamforming together with real-time target tracking have become increasingly vital for applications such as unmanned aerial vehicle (UAV) surveillance, vehicular sensing and near-field radar operations [1], [2]. Prior work has demonstrated the practical advantages of 3-D beam sweeping for spatial coverage and tracking performance [3], and recent surveys categorize traditional sweeping and tracking strategies into two broad classes: blind scanning without prior information and directional narrowing scanning using prior information [4]. However, in realistic deployments, the combination of receiver access points (Rx) and transmitter access points (Tx) yields a four-dimensional (4-D) angular domain: azimuth and elevation for both angle-of-arrival (AoA) and angle-of-departure (AoD). Direct exhaustive search over this 4-D space incurs exponential complexity growth [5].

When a dominant line-of-sight (LoS) component is present, it manifests as an approximately rank-1 contribution on the array manifold, which tends to mask weak target echoes and degrades both detection probability and parameter estimation accuracy. Numerical studies have shown that LoS strength and network density critically affect sweeping efficiency and tracking precision [6]. From a signal model perspective, the effective channel at the receiver can be expressed as the superposition of a dominant LoS term, and the scattering target returns. Consequently, an effective LoS suppression strategy should not only estimate the LoS complex amplitude and phase, but also remove that contribution during beam sweeping so as to unmask and recover the weak target echo.

A range of techniques have been pursued for 3-D AoA/AoD estimation and beam tracking. Low-complexity subspace-based estimators (such as multiple signal classification (MUSIC)), grid searches, learning-based schemes and coordinate-descent approaches are all part of the literature landscape [7]. For instance, Li et al. [8] proposed a scanning and covariance reconstruction procedure that samples a predefined angle set to recover AoA/AoD information, although effective in principle, its accuracy is inherently constrained by grid resolution, resulting in large computational and latency costs as the number of grid points increases. To reduce complexity and improve real-time performance of the system, Shu et al. [9] proposed a root-MUSIC-HDAPA and a two-step phase alignment scheme based on root-MUSIC to achieve high-resolution AoD estimation under the hybrid analog and digital (HAD) architecture. However, this method causes phase ambiguity and requires additional phase alignment or time block overhead, increasing the delay of beam sweeping and limits the real-time tracking ability of fast moving targets. In scenarios where single beam/link analog beamformers are deployed, MUSIC requires to reconstruct the covariance matrix [10], involving diagonal loading method with high complexity. Moreover, deep neural network (DNN) methods have been introduced to reduce online computational burden [11], but their real-time advantages depend strongly on the represen-

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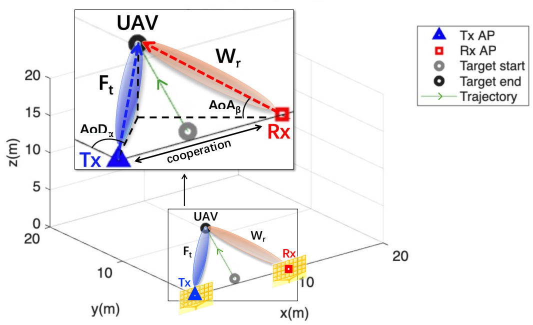


Fig. 1. UAV trajectory diagram in 3-D beamforming scene.

tativeness and generalization of offline training data [12]. To remove the dependence on training data, Cao et al. [13] applied the coordinate descent algorithm to the 3-D cell-free massive multiple input and multiple output (MIMO) integrated sensing and communication (ISAC) system, but ignored the LoS path (between Rx and Tx) interference. Collectively, there is still a lack of a complete process with low complexity, real-time feasibility and explicit suppression of LoS components.

In view of the near-field 3-D beamforming UAV target real-time tracking application requires high number of search dimensions, a single-beam/link analog beamforming and tracking system (Fig.1) featuring LoS path interference suppression and cell-free Rx-Tx cooperation based on low-complexity coordinate descent algorithm and neighborhood searching is designed. MIMO uniform rectangular array of antennas is utilized to be compatible with millimeter-wave (mm-Wave) ISAC systems.

II. Signal Model and Problem Definition

A. 3-D beamforming model with LoS path interference

The 3-D beamforming model with LoS path interference is considered as:

$$\widetilde{Y}[k]=\sum_{i=0}^{Q}\mathbf{W}_r^H\mathbf{H_i}[k]\,\mathbf{F}_t\widetilde{X}[k]+\mathbf{W}_r^H\mathbf{N}[k]=\widetilde{H}[k]\,\widetilde{X}[k]+\widetilde{N}[k] \tag{1}$$

where Q is the quantity of targets, k is the slot index, $i=0$ th path denotes LoS path, $\mathbf{W}_r$ and $\mathbf{F}_t$ are the receiving and transmitting beamforming column vectors respectively and $\mathbf{N}$ is the additive white Gaussian noise (AWGN) with zero-mean and variance $\sigma_{\mathbf{w}}^2$ with $\mathbf{N}[k]\sim\mathcal{N}\left(\mathbf{0},\sigma_{\mathbf{w}}^2\right)$. Suppose $i=2\cdots Q$ th targets are jamming targets, then according to Wu et al [14], the SNR of the proposed sensing system can be calculated as:

$$\gamma=\frac{\left|\mathbf{W}_r^H\mathbf{H}_1[k]\,\mathbf{F}_t\right|^2}{\left|\mathbf{W}_r^H\mathbf{H}_0[k]\,\mathbf{F}_t\right|^2+\sum_{i=2}^{Q}\left|\mathbf{W}_r^H\mathbf{H}_i[k]\,\mathbf{F}_t\right|^2+\sigma_{\mathbf{w}}^2} \tag{2}$$

The steering vectors of 3-D beamforming model are [15]:

$$\begin{aligned}\mathbf{a}_{r,y}\left(\beta_i[k]\right)&=\frac{1}{\sqrt{N_{r,y}}}\left[1\quad e^{j\pi 1\sin\beta_i[k]}\quad\cdots\quad e^{j\pi\left(N_{r,y}-1\right)\sin\beta_i[k]}\right]^T\\ \mathbf{a}_{r,x}\left(\mathrm{a}_i[k],\beta_i[k]\right)&=\frac{1}{\sqrt{N_{r,x}}}\left[1\quad\cdots\quad e^{j\pi\left(N_{r,x}-1\right)\sin\mathrm{a}_i[k]\cos\beta_i[k]}\right]^T\end{aligned} \tag{3}$$

where $\mathrm{a}_i[k]$ and $\beta_i[k]$ are the true azimuth and elevation angle of the UAV target in k th slot channel respectively. The channel gain is calculated as [16]:

$$\begin{aligned}\widetilde{H}[k]&=\sum_{i=0}^{Q}\alpha_i[k]\cdot\underbrace{\mathbf{W}_r^H\mathbf{a}_r}_{\widetilde{a}_r[k]}\cdot\underbrace{\mathbf{a}_t^H\mathbf{F}_t}_{\widetilde{a}_t[k]}\\&=\sum_{i=0}^{Q}\alpha_i[k]\left[\mathbf{W}_{r,y}\otimes\mathbf{W}_{r,x}\right]^H\\&\quad\left[\mathbf{a}_{r,y}\left(\beta_i[k]\right)\otimes\mathbf{a}_{r,x}\left(\mathrm{a}_i[k],\beta_i[k]\right)\right]\\&\quad\cdot\left[\mathbf{a}_{t,y}\left(\beta_i'[k]\right)\otimes\mathbf{a}_{t,x}\left(\mathrm{a}_i'[k],\beta_i'[k]\right)\right]^H\left[\mathbf{F}_{t,y}\otimes\mathbf{F}_{t,x}\right]\end{aligned} \tag{4}$$

The path gain containing radar cross-section (RCS) information is a first-order Markov model [17]: $\alpha_i[k+1]=\rho\cdot\alpha_i[k]+n[k]$ where $n\sim\mathcal{CN}\left(\mu=0,\sigma^2\right)$.

Define the channel of the virtual antenna array is:

$$\mathbf{H_i}[k]=\alpha_i[k]\,\mathbf{A}_i[k] \tag{5}$$

where $\mathbf{A}_i[k]=\mathbf{a}_r\left(\mathrm{a}_0[k],\beta_0[k]\right)\mathbf{a}_t^H\left(\mathrm{a}_0'[k],\beta_0'[k]\right)$, $\forall i=0,1,\cdots,Q$.

Then the channel gain is expressed as:

$$\begin{aligned}\widetilde{H}[k]&=\sum_{i=0}^{Q}\mathbf{W}_r^H\left(AoA[k]\right)\mathbf{H}_i[k]\,\mathbf{F}_t\left(AoD[k]\right)\\&=\mathbf{W}_r^H\left(AoA[k]\right)\mathbf{H}_0[k]\,\mathbf{F}_t\left(AoD[k]\right)\\&\quad+\sum_{i=1}^{Q}\mathbf{W}_r^H\left(AoA[k]\right)\mathbf{H}_i[k]\,\mathbf{F}_t\left(AoD[k]\right)\\&=\underbrace{\alpha_0[k]\,\mathbf{W}_r^H\left(AoA[k]\right)\mathbf{A}_0[k]\,\mathbf{F}_t\left(AoD[k]\right)}_{\text{LoS Path}}\\&\quad+\sum_{i=1}^{Q}\alpha_i[k]\,\mathbf{W}_r^H\left(AoA[k]\right)\mathbf{A}_i[k]\,\mathbf{F}_t\left(AoD[k]\right)\end{aligned} \tag{6}$$

where $AoA[k]$ denotes $\left(AoA_{\mathrm{a}}[k],AoA_{\beta}[k]\right)$ and $AoD[k]$ denotes $\left(AoD_{\mathrm{a}}[k],AoD_{\beta}[k]\right)$ and the azimuth and elevation angle of the AoA and AoD of the beamformer are all certain in every slot k, then $\mathbf{W}_r$ and $\mathbf{F}_t$ are known by the system.

As to the LoS path on the line between Rx and Tx, $\alpha_0[k]$, $\mathrm{a}_0[k]=-\frac{\pi}{2}$, $\beta_0[k]=0$; $\mathrm{a}_0'[k]=\frac{\pi}{2}$, $\beta_0'[k]=0$, then:

$$\begin{aligned}&\mathbf{A}_0[k]_{N_{r,y}N_{r,x}\times N_{t,y}N_{t,x}}\\&=\left[\mathbf{a}_{r,y}\left(\beta_0[k]\right)\otimes\mathbf{a}_{r,x}\left(\mathrm{a}_0[k],\beta_0[k]\right)\right]\\&\quad\cdot\left[\mathbf{a}_{t,y}\left(\beta_0'[k]\right)\otimes\mathbf{a}_{t,x}\left(\mathrm{a}_0'[k],\beta_0'[k]\right)\right]^H\\&=\frac{1}{\sqrt{N_{r,y}N_{r,x}N_{t,y}N_{t,x}}}\begin{bmatrix}1&-1&1&\cdots\\-1&1&-1&\cdots\\&&&\\1&-1&1&\vdots\\\vdots&\vdots&\cdots&\ddots\end{bmatrix}\end{aligned} \tag{7}$$

Eq. (7) holds when $N_{r,x}$ and $N_{t,x}$ are even and the directions of the antenna arrays of Tx and Rx are ideally parallel. For other Tx and Rx layouts, $\mathbf{A}_0[k]$ remains a constant rank-1 matrix if the vibration of the antennas is ignored. The $N_{r,y}N_{r,x}\times N_{t,y}N_{t,x}$ matrix $\mathbf{A}_0$ should be measured and calibrated prior to practical experiments.

Algorithm 1 LoS-suppressing beam sweeping algorithm

Output: AoA^{*}, AoD^{*}, $\hat{\alpha}_0$, $\hat{\alpha}_1$

Initialization:

$k=1$, $AoA\left[k\right]=\left(-\frac{\pi}{2},0\right)$, $AoD\left[k\right]=\left(\frac{\pi}{2},0\right)$

by (4) calculate $\widetilde{H}\left[k\right]$

$\hat{\alpha}_0\left[k\right]=\frac{\widetilde{H}[k]}{\mathbf{W}_r^H\left(-\frac{\pi}{2},0\right)\mathbf{A}_0\mathbf{F}_t\left(\frac{\pi}{2},0\right)}$

$k=2$

$\max\limits_{AoA[k],AoD[k]}|\widetilde{H}_1[k]|=|\widetilde{H}[k]-\hat{\alpha}_0[k-1]\mathbf{W}_r^H(AoA[k])\mathbf{A}_0\mathbf{F}_t(AoD[k])|$

$AoA^{*}=AoA\left[k\right]$, $AoD^{*}=AoD\left[k\right]$

$\hat{\alpha}_1\left[k\right]=\frac{\widetilde{H}_1[k]}{\mathbf{W}_r^H(AoA[k])\mathbf{A}_1(AoA[k],AoD[k])\,\mathbf{F}_t(AoD[k])}$

Procedure:

for $k=3$ to k_{max} **do**

  if k is odd **then**

    $AoA\left[k\right]=\left(-\frac{\pi}{2},0\right)$, $AoD\left[k\right]=\left(\frac{\pi}{2},0\right)$

    by (4) calculate $\widetilde{H}\left[k\right]$

    $\widetilde{H}_0\left[k\right]=\widetilde{H}\left[k\right]-\hat{\alpha}_1\left[k-1\right]\mathbf{W}_r^H\left(-\frac{\pi}{2},0\right)$ $\cdot\mathbf{A}_1\left(AoA\left[k-1\right],AoD\left[k-1\right]\right)\mathbf{F}_t\left(\frac{\pi}{2},0\right)$

    $\hat{\alpha}_0\left[k\right]=\frac{\widetilde{H}_0[k]}{\mathbf{W}_r^H\left(-\frac{\pi}{2},0\right)\mathbf{A}_0\mathbf{F}_t\left(\frac{\pi}{2},0\right)}$

  end if

  if k is even **then**

    $\max\limits_{AoA[k],AoD[k]}|\widetilde{H}_1[k]|=|\widetilde{H}[k]-\hat{\alpha}_0[k-1]\mathbf{W}_r^H(AoA[k])\mathbf{A}_0\mathbf{F}_t(AoD[k])|$

    $AoA^{*}=AoA\left[k\right]$, $AoD^{*}=AoD\left[k\right]$

    $\hat{\alpha}_1\left[k\right]=\frac{\widetilde{H}_1[k]}{\mathbf{W}_r^H(AoA[k])\mathbf{A}_1(AoA[k],AoD[k])\mathbf{F}_t(AoD[k])}$

  end if

  $k=k+1$

end for

Algorithm 2 LoS-suppressing beam tracking algorithm

Input: AoA^{*}, AoD^{*}, $\hat{\alpha}_0$, $\hat{\alpha}_1$, L_{a}, L_{β}, θ_{res}, γ_{th}

Output: AoA^{*}, AoD^{*}, $\hat{\alpha}_0$, $\hat{\alpha}_1$

Initialization:

do **Algorithm 1**

$AoA^{0}=AoA^{*}$, $AoD^{0}=AoD^{*}$, $\gamma^{*}=0$

Procedure:

while $k\leq K_{max}$ **do**

  if $|\widetilde{H}_1\left[k\right]|<\gamma_{th}$ **then**

    do **Algorithm 1**

  end if

  if $1\leq k \bmod 2L_{\mathrm{a}}+2L_{\beta}\leq L_{\mathrm{a}}$ **then**

$AoA_{\mathrm{a}}\left[k\right]=AoA_{\mathrm{a}}^{0}+\lfloor\frac{k \bmod 2L_{\mathrm{a}}+2L_{\beta}}{2}\rfloor\left(-1\right)^{k \bmod 2L_{\mathrm{a}}+2L_{\beta}}\cdot\theta_{res}$

    by (4) and (9) calculate $\widetilde{H}_1\left[k\right]$

    if $|\widetilde{H}_1\left[k\right]|\geq|\gamma^{*}|$ **then**

      $AoA_{\mathrm{a}}^{*}=AoA_{\mathrm{a}}\left[k\right]$, $\gamma^{*}=\widetilde{H}_1\left[k\right]$

    end if

    if $k \bmod 2L_{\mathrm{a}}+2L_{\beta}=L_{\mathrm{a}}$ **then**

      $AoA_{\mathrm{a}}\left[k\right]=AoA_{\mathrm{a}}^{*}$

      $\hat{\alpha}_1\left[k\right]=\frac{\gamma^{*}}{\mathbf{W}_r^H(AoA^{*})\mathbf{A}_1(AoA^{*},AoD^{*})\mathbf{F}_t(AoD^{*})}$

      $AoA_{\mathrm{a}}^{0}=AoA_{\mathrm{a}}^{*}$, $\gamma^{*}=0$

    end if

  end if

  Similarly, optimize $AoA_{\beta}\left[k\right]$, $AoD_{\mathrm{a}}\left[k\right]$, $AoD_{\beta}\left[k\right]$ according to $k \bmod 2L_{\mathrm{a}}+2L_{\beta}$ and update $\hat{\alpha}_1\left[k\right]$.

  if $k \bmod 2L_{\mathrm{a}}+2L_{\beta}=0$ **then**

    $AoA\left[k\right]=\left(-\frac{\pi}{2},0\right)$, $AoD\left[k\right]=\left(\frac{\pi}{2},0\right)$

    by (4) calculate $\widetilde{H}\left[k\right]$

    $\widetilde{H}_0[k]=\widetilde{H}[k]-\hat{\alpha}_1[k-1]\mathbf{W}_r^H\left(-\frac{\pi}{2},0\right)\mathbf{A}_1\left(AoA^{*},AoD^{*}\right)\mathbf{F}_t\left(\frac{\pi}{2},0\right)$

    $\hat{\alpha}_0\left[k\right]=\frac{\widetilde{H}_0[k]}{\mathbf{W}_r^H\left(-\frac{\pi}{2},0\right)\mathbf{A}_0\mathbf{F}_t\left(\frac{\pi}{2},0\right)}$

  end if

  $k=k+1$

end while

B. LoS path interference problem definition

Omitting jamming targets, i.e. $Q=1$ and suppose the power of AWGN $\sigma_{\mathbf{w}}^2$ is constant, the tracking process is equivalent to the maximization of the target path channel gain component:

$$\max_{AoA[k],AoD[k]} |\widetilde{H}\left[k\right]-\hat{\alpha}_0[k]\mathbf{W}_r^H\left(AoA\left[k\right]\right)\mathbf{A}_0\mathbf{F}_t\left(AoD\left[k\right]\right)| \quad (8)$$

Since $\mathbf{W}_r\left(AoA\left[k\right]\right)$, $\mathbf{F}_t\left(AoD\left[k\right]\right)$ and $\mathbf{A}_0$ are known to the cooperation system, it is essential to obtain the estimation of $\alpha_0\left[k\right]$.

The optimization object in (8) is the estimation of the channel gain of the target path $i=1$, hence denote it as:

$$\widetilde{H}_1\left[k\right]=\widetilde{H}\left[k\right]-\hat{\alpha}_0\mathbf{W}_r^H\left(AoA\left[k\right]\right)\mathbf{A}_0\mathbf{F}_t\left(AoD\left[k\right]\right) \quad (9)$$

III. LoS path interference suppressing and tracking method

A. LoS-suppressing beam sweeping method

The proposed LoS path interference suppression beam sweeping algorithm (**Algorithm 1**) alternately calibrates $\hat{\alpha}_0\left[k\right]$ and $\hat{\alpha}_1\left[k\right]$ in iterations by maximizing the target path channel gain component $\widetilde{H}_1\left[k\right]$. Denote AoA^{*} and AoD^{*} as the optimized AoA and AoD of the receiving and transmitting beamformer provided by an exhaustive search of the 4-D code book.

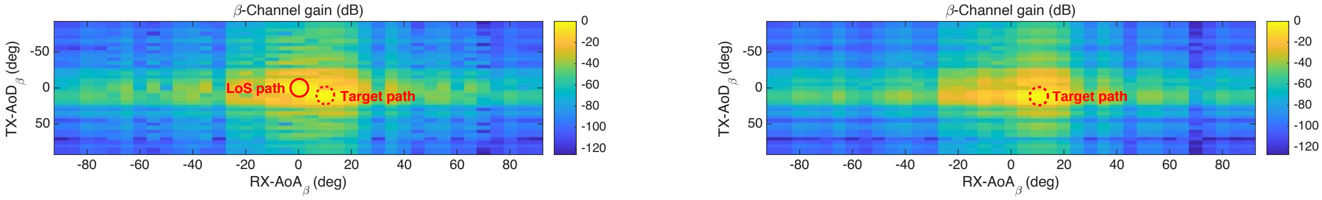
B. LoS-suppressing beam tracking method

The proposed LoS path interference suppression beam tracking method iterates to maximize $\widetilde{H}_1\left[k\right]$ in 4 dimensions of azimuth and elevation AoA and AoD with low complexity, as shown in **Algorithm 2**. In each end of the iteration, the receiving and transmitting beams are pointed in the direction of LoS path to measure the LoS path channel gain component $\widetilde{H}_0\left[k\right]$ to calibrate $\hat{\alpha}_0$. Denote L_{a} and L_{β} as the length of the neighborhood searching window of the azimuth and elevation angle, respectively, and θ_{res} is the grain resolution of the code book of the beamformer. γ_{th} is the minimum acceptable threshold channel gain. The proposed LoS-suppressing beam tracking algorithm is based on coordinate descent algorithm and neighborhood search for their low complexity in real-time scene. The complexity of **Algorithm 2** in each tracking period is $\mathcal{O}\left(2L_{\mathrm{a}}+2L_{\beta}\right)$ [13]. This complexity is significantly smaller than the complexity of beam sweeping and exhaustive search in the 4-D neighborhood, which is $\mathcal{O}\left(L_{\mathrm{a}}^2\cdot L_{\beta}^2\right)$. Note that $\left(AoA_{\mathrm{a}}^{*},AoA_{\beta}^{*}\right)=AoA^{*}$ and $\left(AoD_{\mathrm{a}}^{*},AoD_{\beta}^{*}\right)=AoD^{*}$.

IV. Simulation results

A. Simulation settings

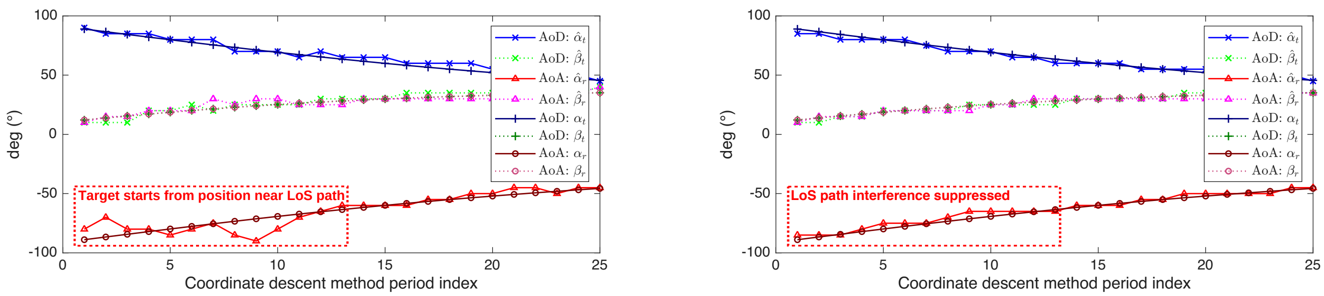
It is found in the simulation that the continuous tracking probability of the coordinate descent algorithm when the target departs from a position close to the LoS path is significantly lower than other circumstances, including when the target trajectory ends at a near-LoS position, without utilization of any LoS path interference suppression measures. Hence, this section presents the scene when the starting position of the UAV target is above the LoS path and close to it, as is showcased in Fig.1. Consequently, it is significant to design a LoS path interference suppression beam sweeping algorithm such as **Algorithm 1** in the initialization stage. The target



(a) Channel gain in LoS path influenced 3-D channel

(b) Channel gain with application of **Algorithm 1**

Fig. 2. Channel gain spectrum of elevation angle domain.



(a) LoS path interfered 3-D time-varying channel

(b) LoS path interfered channel applying **Algorithm 2**

Fig. 3. Beam tracking process of the near-LoS departing target when SNR is -10 dB.

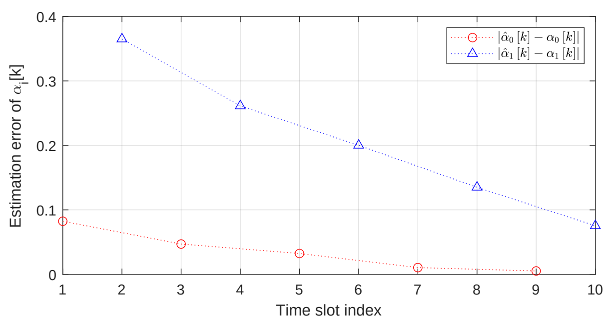
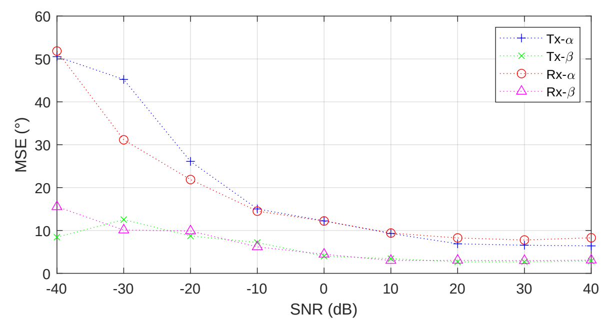
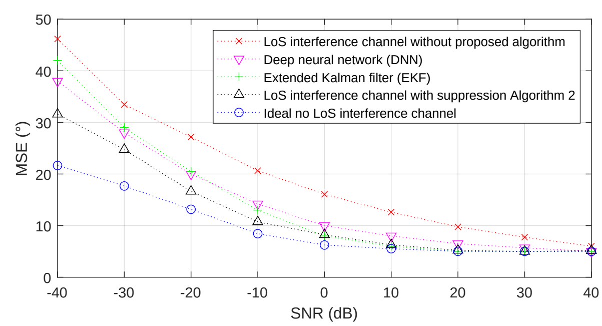


Fig. 4. Convergence of parameter estimation in **Algorithm 1**.



(a) Performance of **Algorithm 2**



(b) Performance comparison

Fig. 5. MSE-SNR performance of AoA and AoD estimation.

path channel in the starting (near-LoS) position is slightly different to the LoS path in elevation angle domain i.e. AoA_β and AoD_β. Code book with $\theta_{res} = 5°$ resolution is used in the beamformer of all access points (APs). The transmitting and receiving APs are all MIMO uniform rectangular array (URA) with 16×8 antennas in $x - y$ axis. The initial LoS path gain is set as 5 times of the initial target path gain to adapt to the mm-Wave radiofrequency environment. Each trajectory consists of 500 points of positions.

B. Beam sweeping results

Assume that the target holds still at the starting position close to the LoS path during the beam sweeping process to evaluate the performance of **Algorithm 1**. Fig.2(a) presents the channel gain $\widetilde{H}[k]$ spectrum of elevation angle β-domain in beam sweeping when SNR is 20. The identification of the target with peak detection methods or neural networks is hindered by the LoS path interference. The LoS path interference component attenuates exponentially when the pair of beams leaves the LoS path. **Algorithm 1** eliminates the LoS path component effectively as is exhibited in Fig.2(b).

Fig.4 displays the convergence of the proposed **Algorithm 1**. The absolute value of the estimation bias of $\alpha_i\left[k\right]$ decreases with the growth of the iteration times.

C. Beam tracking performance

Fig.3(a) presents a successful tracking process of the UAV target starting from a near-LoS position and ending at a random position without application of any LoS-suppressing methods. In the initial tracking periods, the AoA and AoD estimations are quite inaccurate due to LoS path interference and noise. The LoS path interference decays apparently with the target leaving the near-LoS positions. In 20 Monte Carlo tests, the cooperating APs lose tracking of the target with considerable probability without the application of **Algorithm 2**, especially when SNR is low. **Algorithm 2** prominently improves the tracking probability and alleviates the interference of the LoS path component, as is shown in Fig.3(b).

Fig.5(a) presents the MSE to SNR performance of **Algorithm 2** in the estimation of four-dimensions of angles and the average generates the performance curve in Fig.5(b). Fig.5(b) shows the significant improvement of **Algorithm 2** in the performance of the AoA and AoD estimation. The actual MSE of LoS interfered channel without application of **Algorithm 2** is much higher than that shown in Fig.5(b) because unsuccessful tracking processes are discarded, while there is almost no unsuccessful tracking process in the 20 times of Monte Carlo test with application of **Algorithm 2**. Moreover, Fig.5(b) clearly shows the improved MSE performance of the proposed **Algorithm 2** over traditional methods such as EKF and DNN when LoS path interference exists and SNR is low.

V. Conclusion

This paper proposes a low-complexity line-of-sight (LoS) path interference suppression method compatible with MIMO antennas for real-time 3-D beamforming and tracking in UAV monitoring applications. The proposed approach employs a coordinate descent algorithm with neighborhood search to alternately estimate and suppress the dominant LoS component during target tracking. The proposed method achieves satisfactory tracking accuracy in near-LoS scenarios. Simulation results demonstrate significant MSE reduction and enhanced robustness under low SNR conditions, with the proposed method maintaining high tracking probability even when the target is close to the LoS path. Without the proposed method, tracking fails with considerable probability in low SNR environments, while the proposed solution achieves near-optimal performance with significantly reduced computational complexity. The proposed method outperforms traditional methods such as EKF and DNN and demonstrates great potential for practical deployment in integrated sensing and communication (ISAC) systems and radar applications where LoS interference is dominant. Future work will explore extensions to multi-target scenarios and adaptive parameter optimization for varying environmental conditions.

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